

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 13A.5

RELATING ENTROPY TO EQUILIBRIUM CONSTANTS

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	13: CHEMICAL EQUILIBRIA
SUB-TOPIC:	13A.5: Relating Entropy to Equilibrium Constants
TOTAL QUESTIONS:	3 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	6 Marks
EXAMINER FAQs:	3 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 13A.5

Q1. 1. The total entropy change, S_{total} , for a reaction at 298 K is $+45.0 \text{ J K}^{-1} \text{ mol}^{-1}$. Calculate the value of the equilibrium constant, K , for this reaction at 298 K. ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$)

[3 Marks]
WCH14/01/Jan23/Q6

Q2. 2. State the relationship between S_{total} and the position of equilibrium when S_{total} is very large and negative.

[2 Marks]
WCH14/01/Jun22/Q7

Q3. 3. Which equation correctly relates total entropy change to the equilibrium constant?

[1 Marks]
WCH14/01/Oct21/Q6

- A. $S_{\text{total}} = -R \ln K$
- B. $S_{\text{total}} = R \ln K$
- C. $K = R \ln S_{\text{total}}$
- D. $S_{\text{total}} = e^{(R/K)}$

SECTION B: WORKED MARK SCHEME — 13A.5

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. The total entropy change, S_{total} , for a reaction at 298 K is $+45.0 \text{ J K}^{-1} \text{ mol}^{-1}$. Calculate the value of the equilibrium constant, K , for this reaction at 298 K. ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$) 1. $S_{\text{total}} = R \ln K$ 2. $\ln K = 45.0 / 8.31 = 5.415$ 3. $K = e^{5.415} = 225$	[3]
Q2	Q2: 2. State the relationship between S_{total} and the position of equilibrium when S_{total} is very large and negative. 1. When S_{total} is large and negative, $\ln K$ is large and negative. 2. K is very small, so the equilibrium lies very far to the left (reaction effectively does not go).	[2]
Q3	Q3: 3. Which equation correctly relates total entropy change to the equilibrium constant? B	[1]

SECTION C: EMPIRICAL EXAMINER FAQS & COMMON TRAPS — 13A.5**FAQ #1: What are the units for K when calculated from S_{total} ?****Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

What are the units for K when calculated from S_{total} ?

MODEL ANSWER / EXAMINER EXPECTATION:

The K calculated from S_{total} is a dimensionless thermodynamic equilibrium constant. You do not need to provide units for K in this specific context unless explicitly asked to deduce them from the equation.

FAQ #2: How do I handle the gas constant R?**Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

How do I handle the gas constant R?

MODEL ANSWER / EXAMINER EXPECTATION:

Always use $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$. Make sure your S_{total} is in $\text{J K}^{-1} \text{ mol}^{-1}$ before using the equation. If S_{total} is given in $\text{kJ K}^{-1} \text{ mol}^{-1}$, multiply by 1000 first.

FAQ #3: Why do I get a math error on my calculator when finding $\ln K$?**Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

Why do I get a math error on my calculator when finding $\ln K$?

MODEL ANSWER / EXAMINER EXPECTATION:

You might be calculating e to a very large positive or negative power. If $\ln K > 100$, K is huge (reaction goes to completion). If $\ln K < -100$, K is tiny (reaction doesn't happen). Write "very large" or "very small" if the calculator overflows.